

ORIGINAL RESEARCH

Cumulative Benefit With Evolocumab in Patients With No Prior Myocardial Infarction or Stroke in the VESALIUS-CV Study

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ABSTRACT

BACKGROUND Proprotein convertase subtilisin/kexin type 9 inhibition with evolocumab reduced the risk of a first major adverse cardiovascular event (MACE) in patients at high cardiovascular risk with no prior myocardial infarction (MI) or stroke in the VESALIUS-CV (Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke) trial. Recurrent MACE is common and important to patients, clinicians, and health care systems.

OBJECTIVES The purpose of this study was to evaluate the effect of evolocumab on total (first and subsequent) MACE.

METHODS The VESALIUS-CV trial randomized patients with qualifying atherosclerosis or high-risk diabetes, no prior MI or stroke, and a low-density lipoprotein cholesterol ≥ 90 mg/dL on optimized lipid-lowering therapy to evolocumab or placebo. The dual primary endpoints were a composite of coronary heart disease death, MI, ischemic stroke, or ischemia-driven arterial revascularization (4-point [4-P] MACE) and 3-point (3-P) MACE (not including ischemia-driven arterial revascularization). In this prespecified analysis, all dual primary endpoint events (first and subsequent) were analyzed using negative binomial regression models.

RESULTS Among 12,257 patients followed for a median of 4.6 years, there were 1,654 first 4-P MACE and 1,107 subsequent events (67% more) for 2,761 total events. There were 779 first 3-P MACE and 146 subsequent events (19% more) for 925 total events. Evolocumab reduced the rate of first 4-P MACE by 19% (HR: 0.81; 95% CI: 0.73-0.89; $P < 0.0001$), was associated with a reduction of subsequent events by 25% (incidence rate ratio [IRR]: 0.75; 95% CI: 0.61-0.91), and reduced total events by 20% (IRR: 0.80; 95% CI: 0.71-0.90; $P = 0.0002$). Likewise for 3-P MACE, evolocumab reduced the rate of first events by 25% (HR: 0.75; 95% CI: 0.65-0.86; $P < 0.0001$), was associated with a reduction of subsequent events by 37% (IRR: 0.63; 95% CI: 0.42-0.94) and reduced total events by 27% (IRR: 0.73; 95% CI: 0.63-0.86; $P = 0.0001$). Based on annualized incidence rate differences over the full follow-up period, evolocumab was projected to prevent 31 first and 24 subsequent 4-P MACE, for 55 total events per 1,000 patients over 5 years (95% CI: for the total-event difference: 36-74). The corresponding estimates for 3-P MACE were 20 first and 5 subsequent events, for 25 total events per 1,000 patients over 5 years (95% CI: for the total event difference: 15-37). Results were consistent across key subgroups, including by statin intensity and presence of qualifying atherosclerosis.

CONCLUSIONS A substantial proportion of patients experiencing a cardiovascular event had more than 1 event. The addition of evolocumab reduced the total number of MACE, providing support for the role of intensive low-density lipoprotein cholesterol lowering to prevent first and subsequent MACE in patients at high cardiovascular risk and no prior MI or stroke. (Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke [VESALIUS-CV]; [NCT03872401](#)) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS

3-P = 3-point

4-P = 4-point

ASCVD = atherosclerotic
cardiovascular disease

CHD = coronary heart disease

IDR = ischemia-driven arterial
revascularizationLDL-C = low-density
lipoprotein cholesterolMACE = major adverse
cardiovascular event

MI = myocardial infarction

PCSK9 = proprotein
convertase subtilisin/kexin
type 9

Randomized clinical trials have traditionally used time-to-first-event analyses to evaluate treatment efficacy, prioritizing simplicity and statistical efficiency; however, this approach captures only the initial manifestation of the designated endpoint and fails to account for the full burden of recurrent adverse events that frequently occur in high-risk populations.^{1,2} Recurrent events contribute substantially to impaired quality of life, increased health care utilization, and overall societal costs.

In recent years, there has been growing recognition of the importance of total (first and subsequent) event analyses, with several large-scale cardiovascular (CV) trials demonstrating that treatment effects may be

more pronounced when total events are considered rather than only first events, and eventually showing incidences of subsequent events surpassing the incidences of the first events.³ Accordingly, there is an increasing emphasis on the importance of considering total event analyses, alongside traditional time-to-first-event approaches, particularly in high-risk populations, to better reflect the cumulative burden of disease and more closely align with patient-centered outcomes.⁴

In the VESALIUS-CV (Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke) trial, the proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor evolocumab significantly reduced the rate of first CV events for the dual primary endpoints in time-to-first-event analyses in a population with atherosclerosis and/or high-risk diabetes without a previous myocardial infarction (MI) or stroke.⁵ In a prespecified analysis of VESALIUS-CV, we evaluated the

effect of evolocumab on total events, including both first and subsequent CV events. By integrating recurrent event analyses, we sought to provide a more comprehensive understanding of clinical efficacy of evolocumab and its impact on the overall burden of CV disease.

METHODS

STUDY POPULATION. The study design and primary results of the VESALIUS-CV trial have been published previously.^{5,6} VESALIUS-CV was a randomized, placebo-controlled, multinational trial of subcutaneously administered evolocumab (140 mg every 2 weeks) in patients without a history of MI or stroke who fulfilled at least 1 of the following 2 risk profiles: qualifying atherosclerosis or high-risk diabetes. Trial participants were required to have a low-density lipoprotein cholesterol (LDL-C) level ≥ 90 mg/dL (2.3 mmol/L), a non-high-density lipoprotein cholesterol level ≥ 120 mg/dL (3.1 mmol/L), or an apolipoprotein B level ≥ 80 mg/dL (≥ 1.56 μ mol/L) on a stable, optimized lipid-lowering therapy for at least 2 weeks. Details of the inclusion and exclusion criteria, including definitions of qualifying atherosclerosis and high-risk diabetes, have been published.^{5,6} The study protocol and amendments were approved by the relevant ethics committees for all participating sites.

OUTCOMES. The dual primary endpoints were the composites of coronary heart disease (CHD) death, MI, ischemic stroke, or ischemia-driven arterial revascularization (IDR) (4-point [4-P] major adverse cardiovascular events [MACE]) and 3-point (3-P) MACE (without IDR). A central clinical events committee led by the TIMI Study Group, whose members were unaware of treatment assignment and lipid

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levels, adjudicated all efficacy endpoints. Definitions of the endpoints have been published previously.⁵

STATISTICAL ANALYSES. As part of a prespecified analysis, negative binomial regression models were performed to compare the total (first and subsequent) number of primary and secondary endpoints between patients in the evolocumab and placebo groups. Log of follow-up time was used as an exposure offset in the model, which included randomized treatment arms as well as the randomization stratification factors of LDL-C level at screening (<160 mg/dL [4.14 mmol/L] or ≥160 mg/dL) and geographic region (North America, Europe, or other). The subsequent event analysis excluded patients in whom the first event was a CHD death; all other randomized and alive patients were included. The incidence rate ratios (IRRs) and corresponding 95% CIs for subsequent events were calculated from the negative binomial regression model by excluding the first event from the analysis. The likelihood ratio test was used to assess the overdispersion parameter in the negative binomial model; all models rejected the null hypotheses of a Poisson distribution, confirming that the negative binomial distribution provided a superior model fit. All models demonstrated stable convergence during maximum likelihood estimation, with final iterations achieving a stable log-likelihood for each model. Kaplan-Meier curves were generated to show first events. Mean cumulative function curves were generated to display total events throughout the follow-up where CHD death was included as an event without additional follow-up and non-CHD death resulted in censoring. Mean cumulative function curves did not account for dependence between recurrent events and death.

The absolute rate-difference decomposition of, first, subsequent, and total event rates were calculated in the full randomized cohort using the same arm-specific total follow-up time, such that the total event rate difference equals the sum of the first and subsequent event rate differences.

The absolute effects estimands for events prevented were estimated using 2 methodologies. For the primary approach reporting events prevented at 5 years per 1,000 patients, the annualized difference in incidence rate per 1,000 patient-years of follow-up between the arms was multiplied by 5 to estimate the 5-year time point. For the secondary approach, cumulative events prevented per 1,000 patients were estimated at selected annual follow-up time points. Incidence rate differences between treatment arms were calculated over successive 1-year intervals, and these interval-specific differences were accumulated

to estimate the cumulative number of events prevented through each specific follow-up time point.

A sensitivity analysis was performed excluding: 1) revascularizations of the relevant vascular bed within 3 days after an ischemic event in that vascular bed; and 2) repeat revascularization within the same vascular bed on the same day. A second sensitivity analysis was performed that excluded a second event within 7 days with exclusion of: 1) revascularizations of the relevant vascular bed within 7 days after an ischemic event in that vascular bed; 2) repeat revascularization within the same vascular bed within 7 days; and 3) CHD death within 7 days after an MI.

Sensitivity analyses were also performed with respect to the modeling approach, using the Wei, Lin, and Weissfeld (WLW) method,⁷ which is a marginal model and an extension of survival models based on the Cox proportional hazard where additional events that could have occurred in a subject were evaluated. With the WLW method, each subject is simultaneously at risk from randomization through follow-up for each event, thereby preserving randomization and allowing for treatment effect estimation for events beyond the first event. The first 4 events were considered for 4-P MACE and the first 3 events were considered for 3-P MACE, because few subjects had more than these number of events yielding unstable effect estimates. Because non-CHD death is a terminating event not part of the composite endpoints and thus a competing risk, a Ghosh-Lin model was performed. This model evaluates the marginal mean function of a recurrent event process while adjusting for a dependent terminal event (eg, non-CHD death) that stops the possibility of recurrence.⁸

Efficacy comparisons were performed according to the intention-to-treat principle. Although prespecified, the analyses of subsequent and total events were not alpha controlled; as such, *P* values and CIs for analyses beyond the first event should be considered nominal. Analyses were conducted with Stata version 18.0 (StataCorp LP) or SAS version 9.4 (SAS Institute).

RESULTS

A total of 12,257 patients were randomized and included in the primary analytic cohort. The median length of follow-up was 4.6 years (Q1-Q3: 4.0-5.2 years; maximum 6.1 years). The baseline characteristics are shown for 4-P (Table 1) and 3-P MACE (Supplemental Table 1) according to whether they experienced no, 1, or more than 1 event during follow-up. Baseline characteristics were generally

TABLE 1 Baseline Characteristics in Patients With No, 1, or Multiple 4-P MACE During Follow-Up

	No Events (n = 10,603)	1 Event (n = 1,010)	Multiple Events (n = 644)	P Value for 1 vs Multiple Events
Demographics				
Female	44.1	32.2	33.1	0.704
White	93.3	91.8	91.6	0.897
Age, y	65 (60-71)	67 (61-72)	66 (61-71)	0.169
BMI, kg/m ²	29.5 (26.6-33.4)	29.4 (26.7-33.0)	30.0 (26.7-33.6)	0.126
Weight, kg	84.7 (74.0-96.2)	85.7 (75.4-96.4)	87.5 (76.0-97.7)	0.220
Region				0.105
North America	10.4	14.2	17.9	
Europe	69.4	68.6	65.7	
Latin America	5.0	6.8	5.1	
Asia	15.3	10.4	11.3	
Qualifying disease state				
Any qualifying atherosclerosis	65.7	77.0	77.0	0.922
Coronary artery disease without previous MI	43.7	54.1	53.7	0.895
Cerebrovascular disease without previous stroke	9.9	9.4	10.2	0.573
Peripheral artery disease	16.3	24.5	25.6	0.593
High-risk diabetes	49.2	46.1	49.7	0.159
High-risk diabetes without qualifying atherosclerosis	34.2	23.0	22.8	0.946
Other comorbidities				
History of hypertension	86.8	87.0	91.0	0.013
History of diabetes	58.3	55.6	59.3	0.141
Current smoker	27.9	26.0	26.9	0.711
eGFR, mL/min/1.73 m ²	78.8 (66.0-92.7)	77.5 (62.9-92.7)	78.0 (62.1-92.2)	0.739
Lipid-lowering therapy				
Any lipid-lowering therapy	92.3	89.4	87.7	0.294
High-intensity lipid-lowering therapy	73.1	66.9	66.6	0.894
Any statin	87.5	83.8	81.1	0.156
High-intensity statin	68.9	62.5	60.7	
Moderate- or low-intensity statin	18.6	21.3	20.3	
Ezetimibe use at baseline	19.5	21.1	23.9	0.178
Lipid measures, mg/dL				
LDL cholesterol	122.0 (104.0-149.0)	119.9 (104.0-146.9)	121.2 (103.6-150.0)	0.631
Non-HDL cholesterol	153.0 (129.9-182.1)	150.8 (128.8-180.0)	151.3 (130.7-185.2)	0.412
HDL cholesterol	47.6 (40.0-57.2)	46.0 (39.8-55.3)	46.0 (38.7-54.1)	0.228
Total cholesterol	202.0 (178.0-233.2)	198.0 (177.9-230.0)	200.0 (176.0-232.0)	0.712
Triglycerides	152.0 (109.8-217.9)	156.0 (115.0-212.6)	156.0 (115.1-237.7)	0.241
Apolipoprotein B	101.0 (88.9-121.0)	99.0 (83.0-120.0)	100.0 (89.0-120.9)	0.579

Values are % or median (25th percentile-75th percentile). Pooled randomization groups.
BMI = body mass index; eGFR = estimated glomerular filtration rate; HDL = high-density lipoprotein; LDL = low-density lipoprotein.

similar among patients with one vs multiple events (Table 1, Supplemental Table 1).

Events. During the trial, there were 1,654 first 4-P MACE and 1,107 subsequent events (67% more), for a total of 2,761 events (Supplemental Figure 1). Of the 1,107 additional 4-P MACE, 870 (79%) were IDRs. For 3-P MACE, there were 779 first events and 146 subsequent events (19% more), for a total of 925 3-P MACE (Supplemental Figure 1). For the individual nonfatal endpoints, there were 1,836 total IDR events, including 1,260 first and 576 subsequent

(46% more) events; 421 total MI events, including 378 first and 43 subsequent (11% more) events; and 282 total ischemic stroke events with 259 first and 23 subsequent (9% more) events.

When considering the total number of events, of the 12,257 subjects, 8.2% (n = 1,010) had a single 4-P MACE, 3.3% (n = 405) had 2 events, and 1.9% (n = 239) had 3 or more events. For 3-P MACE, 5.3% (n = 650) had a single 3-P MACE, whereas 1.1% (n = 129) had 2 or more events. The maximum number of events an individual patient experienced was

19 for 4-P MACE and 5 for 3-P MACE. The median time between serial events was 36 days for 4-P MACE and 65 days for 3-P MACE.

Changes in lipid-lowering therapy. With respect to continuation of randomized treatment after a first nonfatal 4-P MACE event, only 1.7% (25/1,508) permanently discontinuing randomized therapy within 1 week of an event. In the trial overall, 570 (4.7%) patients had an intensification of lipid-lowering therapy at the end of study compared with the baseline regimen.^{5,9} Overall during the trial, 294 (2.4%) patients initiated open label PCSK9 inhibition and 557 (4.5%) initiated ezetimibe or bempedoic acid, with numerically higher rates in the placebo arm.⁵ Of those with a first nonfatal 4-P MACE event, 14.7% (222 of 1,507) had an intensification of lipid-lowering therapy after randomization, with numerically higher rates of any intensification in the placebo arm (19.5%, 163 of 834) compared with the evolocumab arm (8.8%, 59 of 673), as well as higher rates of initiation of PCSK9 inhibition (9.7% [81 of 834] in the placebo arm vs 3.4% [23 of 674] in the evolocumab arm) and ezetimibe or bempedoic acid (17.3% [144 of 834] in the placebo arm vs 6.8% [46 of 674] in the evolocumab arm).

Efficacy. As previously reported, evolocumab significantly reduced the rate of a first 4-P MACE by 19%, with 160 fewer first events, compared with the placebo group ($n = 747$ events, 5-year Kaplan-Meier rate 13.4% vs $n = 907$ events, 5-year Kaplan-Meier rate 16.2%; HR: 0.81; 95% CI: 0.73-0.89; $P < 0.0001$) (Figure 1).⁵ Furthermore, evolocumab was associated with a reduction in subsequent 4-P MACE by 25% (IRR: 0.75; 95% CI: 0.61-0.91) and reduced total 4-P MACE by 20% (IRR: 0.80; 95% CI: 0.71-0.90; $P = 0.0002$) (Figure 1). In a sensitivity analysis using the Ghosh-Lin model to account for the competing risk of non-CHD death, results were similar (rate ratio: 0.81; 95% CI: 0.72-0.92; $P < 0.001$). Using the full duration of follow-up, the annualized difference in incidence rates for total 4-P MACE between the evolocumab and placebo treatment arms was 1.09% (95% CI: 0.71%-1.47%). Based on annualized incidence-rate differences over the full follow-up period, evolocumab was projected to prevent 31 first and 24 subsequent 4-P MACE, for 55 total events per 1,000 patients over 5 years (95% CI for the total-event difference: 36-74). Estimates for the cumulative 4-P MACE prevented at specified time points using the secondary method are depicted in Figure 1B.

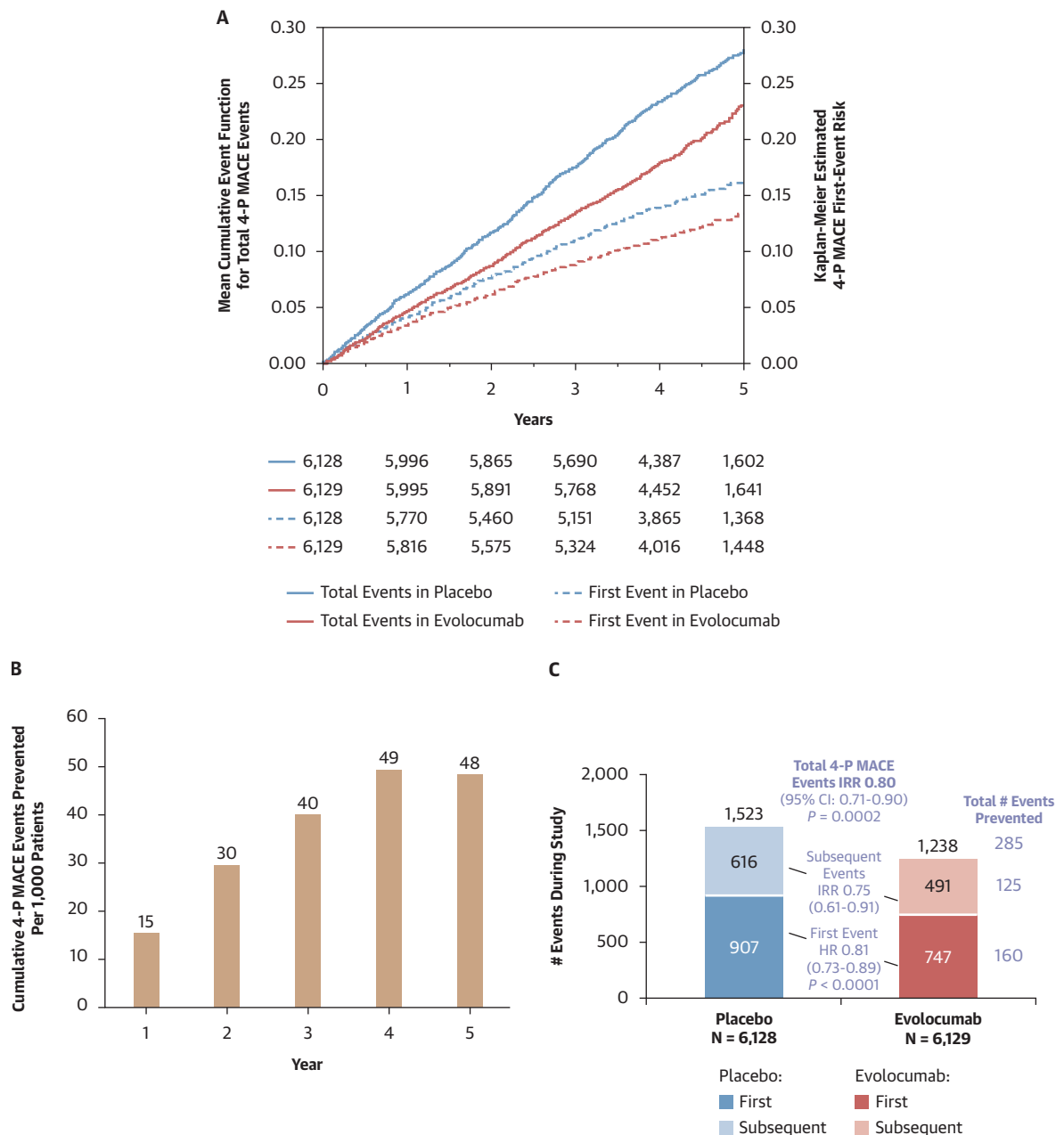
Likewise, as previously reported, evolocumab significantly reduced the rate of a first 3-P MACE by 25%, with 107 fewer first events, compared with the

placebo group ($n = 336$ events, 5-year Kaplan-Meier rate 6.2% vs $n = 443$ events, 8.0%; HR: 0.75; 95% CI: 0.65-0.86; $P < 0.0001$) (Figure 2).⁵ In addition, evolocumab was associated with a reduction in subsequent 3-P MACE by 37% (IRR: 0.63; 95% CI: 0.42-0.94) and reduced total 3-P MACE by 27% (IRR: 0.73; 95% CI: 0.63-0.86; $P = 0.0001$) (Figure 2). In a sensitivity analysis using the Ghosh-Lin model to account for the competing risk of non-CHD death, results were similar (rate ratio: 0.75; 95% CI: 0.64-0.86; $P < 0.001$). Using the full duration of follow-up, the annualized difference in incidence rates for total 3-P MACE between the evolocumab and placebo treatment arms was 0.51% (95% CI: 0.29%-0.73%). Based on annualized incidence rate differences over the full follow-up period, evolocumab was projected to prevent 20 first and 5 subsequent 3-P MACE, for 25 total events per 1,000 patients over 5 years (95% CI: for the total event difference, 15-37). Estimates for the cumulative 3-P MACE prevented at specified time points using the secondary method are depicted in Figure 2B.

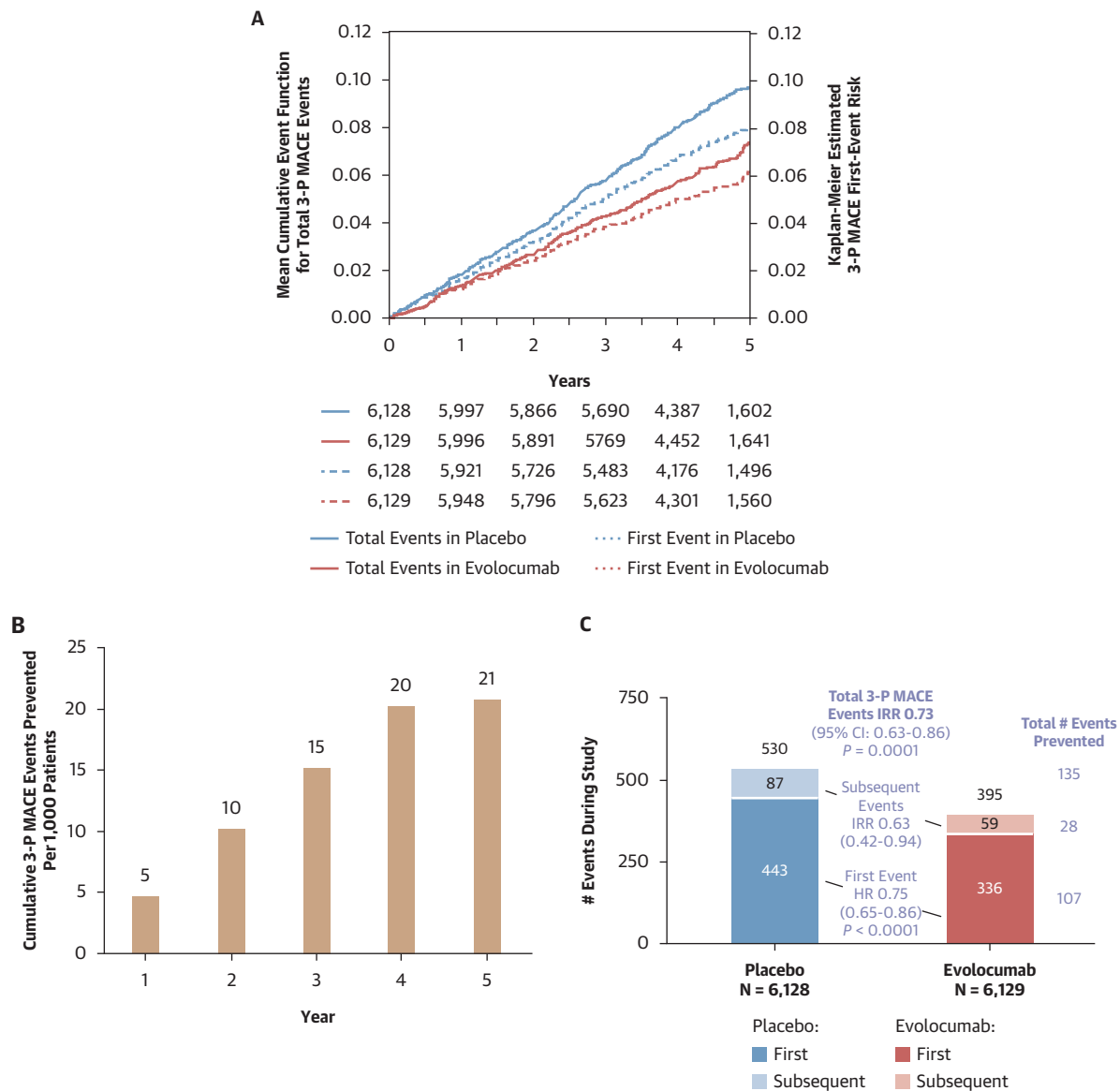
When comparing evolocumab vs placebo, there was a 36% reduction in the total number of MIs (IRR: 0.64; 95% CI: 0.51-0.79; $P < 0.0001$), a 22% reduction in the total number of ischemic strokes (IRR: 0.78; 95% CI: 0.60-1.00; $P = 0.054$), and a 17% reduction in IDRs (IRR: 0.83; 95% CI: 0.73-0.95; $P = 0.005$) (Supplemental Figure 2).

In a sensitivity analysis that excluded revascularizations of the relevant vascular bed within 3 days after an ischemic event and repeat revascularization within the same vascular bed on the same day, evolocumab prevented a total of 228 4-P MACE with a similar treatment effect to that observed in the primary analysis (IRR: 0.82; 95% CI: 0.73-0.92; $P = 0.0005$) (Supplemental Figure 3). In a sensitivity that excluded subsequent events that occurred within 7 days after a preceding event, evolocumab prevented 209 total 4-P MACE with a 17% relative reduction (IRR: 0.83; 95% CI: 0.74-0.93; $P = 0.001$) and 123 total 3-P MACE with a 25% relative reduction (IRR: 0.75; 95% CI: 0.64-0.87; $P < 0.001$) (Supplemental Figure 4).

In sensitivity analyses using the Wei, Lin, and Weissfeld Cox model, evolocumab was associated with a reduction in the occurrence of additional 4-P MACE endpoint events by 30% (HR: 0.70; 95% CI: 0.60-0.82) for second events, and 21% (HR: 0.79; 95% CI: 0.61-1.02) for third events (Figure 3). Similarly, evolocumab was associated with a reduction in the occurrence of additional 3-P MACE by 31% (HR: 0.69; 95% CI: 0.49-0.98) for second events (Figure 3).

FIGURE 1 Total Events Analysis for the 4-P MACE Composite

(A) The mean cumulative event count curves for total (first and subsequent) events (solid lines) per person and Kaplan-Meier estimated event rate curves for first events only (dashed lines) for 4-point (4-P) major adverse cardiovascular events (MACE) (coronary heart disease death, myocardial infarction, ischemic stroke, or ischemia-driven arterial revascularization). (B) Estimates of the cumulative number of 4-P MACE prevented with evolocumab compared with placebo at the indicated year per 1,000 patients using the secondary method for estimating absolute effects. CIs were not calculated. (C) The count of total (first and subsequent) events for 4-P MACE over the total duration of follow-up with the HR from stratified Cox proportional hazards modeling for first events and incidence rate ratios (IRRs) from negative binomial modeling for subsequent and total events, along with the corresponding 95% CIs.

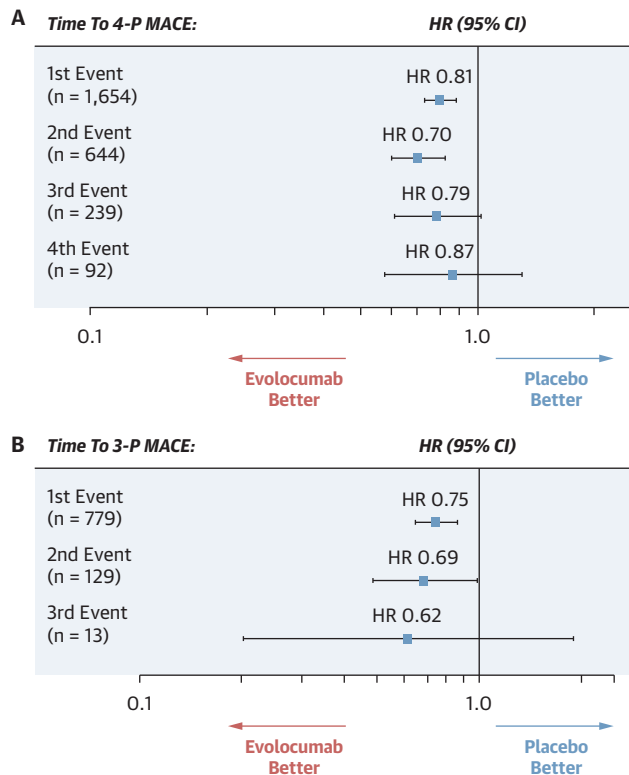
FIGURE 2 Total Events Analysis for the 3-P MACE Composite

(A) The mean cumulative event count curves for total (first and subsequent) events (solid lines) per person and Kaplan-Meier estimated event rate curves for first events only (dashed lines) for 3-point (3-P) MACE (coronary heart disease death, myocardial infarction, or ischemic stroke). (B) Estimates of the cumulative number of 3-P MACE prevented with evolocumab compared with placebo at the indicated year per 1,000 patients using the secondary method for estimating absolute effects. CIs were not calculated. (C) The count of total (first and subsequent) events for 3-P MACE over the total duration of follow-up with the HR from stratified Cox proportional hazards modeling for first events and IRRs from negative binomial modeling for subsequent and total events, along with the corresponding 95% CIs. Abbreviations as in Figure 1.

Total event reductions for 4- and 3-P MACE were consistent across key subgroups, including by age, sex, statin intensity, and the presence of qualifying atherosclerosis or high-risk diabetes without qualifying atherosclerosis (Figure 4).

DISCUSSION

The VESALIUS-CV trial demonstrated that evolocumab reduced the first occurrence of the dual primary endpoints of 4- and 3-P MACE by 19% and 25%,

FIGURE 3 Sensitivity Analysis Using Wei, Lin, and Weissfeld Method

The dual primary endpoints by randomization group as assessed using the Wei, Lin, Weissfeld model in (A) 4-point (4-P) major adverse cardiovascular events (MACE) (coronary heart disease death, myocardial infarction, ischemic stroke, or ischemia-driven arterial revascularization) and (B) 3-point (3-P) MACE (coronary heart disease death, myocardial infarction, or ischemic stroke) comparing the time to the indicated event (eg, time to second event) by randomized treatment. HR, 95% CI, and number of events (n) depicted.

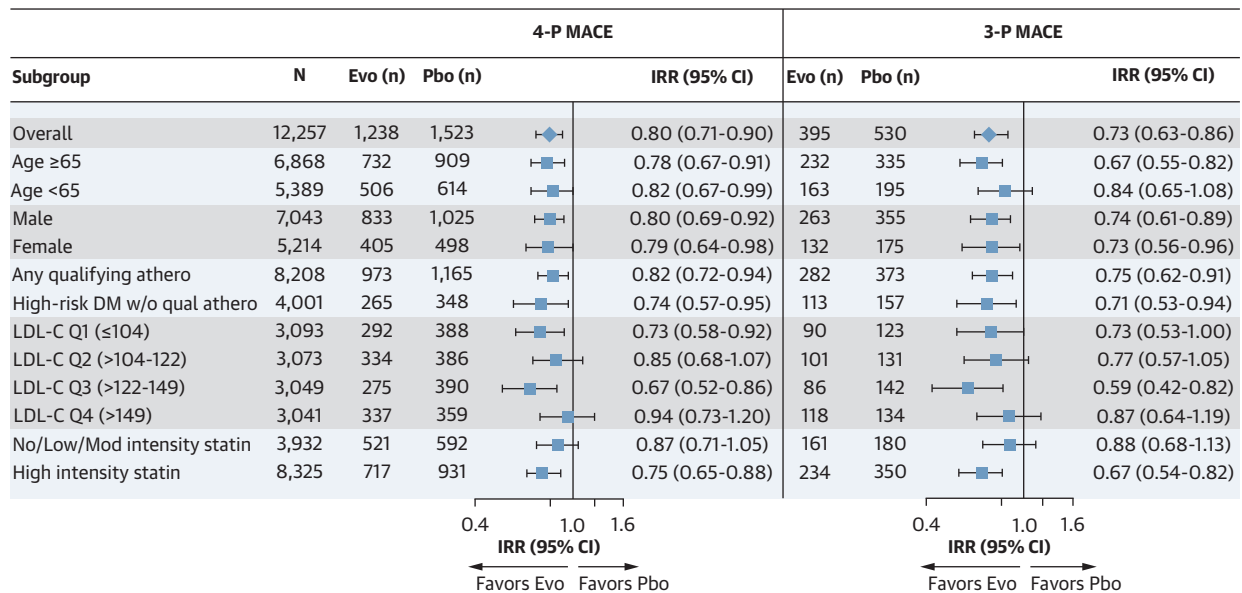
respectively, in patients at high CV risk without a prior MI or stroke.⁵ This prespecified analysis of VESALIUS-CV expands the findings by demonstrating that earlier, intensive LDL-C lowering with evolocumab not only reduces the occurrence of a first major CV event, but also the occurrence of subsequent and total major CV events, emphasizing the importance of sustained, intensive lipid lowering early and across the atherosclerotic cardiovascular disease (ASCVD) continuum.

It is estimated that approximately 612 million people globally currently live with CV diseases, with about 20 million deaths annually—approximately one-third of all global deaths—attributed to CV disease.¹⁰ CV disease, in its various forms, substantially affects quality of life and accounts for trillions of dollars in annual health expenditures.¹¹ Importantly, ASCVD should be viewed as a continuum, spanning

from a high risk for development of disease in the face of risk factors, to the presence of subclinical disease without overt atherothrombotic manifestations, to clinically manifest ASCVD (eg, prior MI or stroke), and finally, to the highest risk with recurrent clinical ASCVD events. Patients can have a biologically important burden of vascular disease even before a first MI or stroke. So, although the occurrence of this first event of MI or stroke likely reflects preexisting vascular disease, it does mark an important transition to a higher risk and higher cost state along the ASCVD continuum. VESALIUS-CV demonstrated that earlier, intensive LDL-C lowering with evolocumab decreases the likelihood that a high-risk patient with risk factors or atherosclerosis crosses that boundary to experience a first event. This current analysis extends the primary aim, quantifying the effect of earlier, intensive LDL-C lowering on preventing total ASCVD events, both first and subsequent, thereby fully capturing more of the cumulative CV event burden across the ASCVD continuum.

Benefits have been observed for prevention of recurrent CV events across a broad range of therapies.¹²⁻¹⁴ Particularly relevant to the current analysis, we and others have reported reductions in the time to a first event after randomization as well as subsequent major adverse CV events with several non-statin lipid-lowering therapies, including ezetimibe, icosapent ethyl, and PCSK9 inhibitors.¹⁵⁻¹⁸ In patients with stable ASCVD with a prior MI, stroke or symptomatic peripheral artery disease, evolocumab resulted in a 15% relative reduction in risk for the first primary endpoint event of CV death, MI, stroke, coronary revascularization, or hospitalization for unstable angina and an 18% relative reduction in the recurrent events over a median of 2.2 years of follow-up.^{16,19} In patients with a recent acute coronary syndrome, alirocumab resulted in a 13% relative reduction in total nonfatal events of MI, stroke, revascularization, hospitalization for unstable angina, and heart failure over a median of 2.8 years of follow-up.¹⁷ In both cases, there were approximately 1.7 to 1.8 times as many total as first events.

The findings described here from VESALIUS-CV now add to this body of knowledge by showing that in patients with atherosclerosis or high-risk diabetes but without a prior MI or stroke, intensive lipid lowering to a median LDL-C of ~45 mg/dL⁵ with evolocumab reduces the occurrence of first and subsequent major CV events. Specifically, over a median follow-up of 4.6 years, evolocumab resulted in a 19% reduction in the first 4-P MACE and a 20% reduction in total 4-P MACE, with nearly double the number of total 4-P MACE prevented, driven in large part by

FIGURE 4 Total Events Analysis for the Dual Primary Endpoints in Key Subgroups

Total (first and subsequent) events analysis using primary analytic approach with negative binomial regression models in 4- and 3-P MACE. N reflects subgroup population size and n, the total events of the indicated endpoint in each subgroup during follow-up. A total of 48 patients were excluded from the disease category subgroup analysis for not meeting either qualifying atherosclerosis or high-risk diabetes criteria. One patient did not have baseline LDL-C values. IRR from a negative binomial model and 95% CI presented. *P* value for interaction >0.05 for each subgroup. Athero = atherosclerosis; DM = diabetes mellitus; LDL-C = low-density lipoprotein cholesterol (in mg/dL); qual = qualifying; other abbreviations as in [Figure 1](#).

relatively high rates of subsequent IDR. Additionally, evolocumab resulted in a 25% relative reduction in the occurrence of a first 3-P MACE and a 27% relative reduction in total 3-P MACE, which excluded revascularization. The findings were consistent in multiple sensitivity analyses and across key subgroups, including those with qualifying atherosclerosis without a prior MI or stroke and those with high-risk diabetes without qualifying atherosclerosis.

Several findings are relevant to consider when putting the current findings in the context of the existing data for the benefit of PCSK9 inhibition for prevention of recurrent events. First, there is very little overlap of the VESALIUS-CV population with that of either FOURIER (Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Subjects with Elevated Risk) or ODYSSEY OUTCOMES (Evaluation of Cardiovascular Outcomes After An Acute Coronary Syndrome During Treatment with Alirocumab), with VESALIUS-CV representing patients on the lower-risk end of the ASCVD continuum and the others including patients with at least one prior major clinical ASCVD event.^{5,19,20} The burden of subsequent events is somewhat less in VESALIUS-CV than FOURIER and ODYSSEY OUTCOMES despite a

longer duration of follow-up in VESALIUS-CV. That likely reflects the very high-risk ASCVD population in the older trials, most of whom had already experienced at least 1 prior major atherothrombotic event. Finally, an apparent greater magnitude of benefit in VESALIUS-CV compared with the other studies may be explained by the longer follow-up as well as the lower risk of the patients, the impact of both of which have been well-described.^{21,22} Therefore, VESALIUS-CV extends the evidence for prevention of first and subsequent CV events with intensive LDL-C lowering with PCSK9 inhibition to a population who had not experienced a prior MI or stroke, followed for a substantially longer period and with an apparent greater magnitude of benefit.

The larger relative proportion of subsequent 4-P MACE (67% more) compared with 3-P MACE (19% more) reflects the high frequency of subsequent IDR events in the VESALIUS-CV population, an event type that may be more influenced by clinical practice patterns than death or nonfatal atherothrombotic events of MI or stroke. Nevertheless, prevention of revascularization is meaningful from a patient-centered and health care expenditure perspective. Furthermore, although recurrent nonfatal MI or

stroke or fatal events are less common than revascularization events, prevention of these important medical events is the primary aim of secondary preventative strategies. Moreover, there was a qualitatively similar relative reduction in subsequent events of 25% for 4-P MACE, which includes revascularization, and 37% for the 3-P MACE composite of CHD death, MI, or ischemic stroke.

STUDY LIMITATIONS. The analysis of additional events following a first event excluded patients where the first event was fatal; because this is a postrandomization event, it could introduce bias. As such, *P* values and CIs for analyses beyond the first event should be considered nominal. The primary analytic method used was a count-based negative binomial model; although this model allows for the inclusion of all possible additional events and incorporates the total follow-up time of a patient, it does not account for the specific timing of the individual events within a subject. As such, a sensitivity analysis was performed using the WLW model, which is an extension of survival models based on the Cox proportional hazard that incorporates the timing of the additional events. In addition, the lower rate of overall mortality observed in the evolocumab arm compared with the placebo arm in the VESALIUS-CV trial introduced the possibility of competing risk bias; a sensitivity analysis using the Ghosh-Lin model to address the competing risk of non-CHD death showed virtually identical results.

CONCLUSIONS

A substantial proportion of patients experiencing a CV event had more than 1 event. The addition of evolocumab reduced the rate of total MACE, providing support for the role of intensive LDL-C lowering to prevent first and subsequent MACE in patients at high CV risk and no prior MI or stroke. By considering subsequent events and therefore the total magnitude of benefit, these observations have relevant implications for patients, providers, and payors, with respect to quality of life and health care system burden and expenditures.

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APPENDIX For a supplemental table and figures, please see the online version of this paper.